

PT ARENA BUOY 46014, CA, USA

WMO: 992500

Lat: 39.231N

Lon: 123.974W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (NAP)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 3	(b) 6.6	(c) 7.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 16.4	(k) 10.6	(l) 15.6	(m) 10.5	(n) 4.3	(o) 70	(p) 0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 2.0	(c) 16.9	(d) N/A	(e) 16.1	(f) N/A	(g) 15.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.4	(p) 330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	14.1	13.1	5.3	18.1	1.3	1.3	4.3	19.0	3.6	19.7	2.8	20.5	1.9	21.4
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	10.7	10.5	10.3	10.4	11.2	11.7	12.0	12.9	13.2	12.6	12.1	10.9
	DBStd	1.68	1.45	1.45	1.32	1.15	1.28	1.05	1.11	1.41	1.47	1.34	1.50	1.64
	HDD10.0	54	9	11	12	8	2	1	0	0	0	1	2	9
	HDD18.3	2477	236	221	249	237	223	199	196	167	153	177	188	231
	CDD10.0	619	32	23	21	21	38	51	62	91	97	82	64	37
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	6.0	5.9	6.3	6.7	6.6	7.0	7.2	5.7	5.4	4.7	5.2	5.5	6.2
(15) Precipitation	PrecAvg	1162	216	184	150	92	50	16	2	4	3	59	124	255
	PrecMax	1847	654	498	408	293	201	77	11	28	23	211	308	595
	PrecMin	743	18	29	19	23	0	0	0	1	0	2	20	52
	PrecStd	346	157	115	114	70	53	20	3	5	6	51	79	163
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	14.6	14.1	14.0	16.8	15.2	15.8	17.9	17.9	16.9	16.5	15.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	13.9	13.8	13.3	13.0	14.6	14.3	14.9	16.7	16.9	15.9	15.5	14.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	13.3	13.0	12.8	12.6	13.4	13.9	14.2	15.7	16.2	15.1	14.9	13.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	12.8	12.5	12.2	12.1	12.8	13.3	13.7	15.0	15.3	14.5	14.3	13.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	MDBR		2.2	2.1	1.9	1.8	1.6	1.5	1.7	1.9	2.0	2.0	2.2	2.3
	5% DB	MCDBR	2.3	2.4	2.2	2.3	2.4	2.0	2.0	2.6	2.5	2.7	2.4	2.4
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.327	0.333	0.335	0.362	0.360	0.354	0.357	0.347	0.339	0.336	0.320	0.328
	taud		2.552	2.537	2.553	2.436	2.483	2.502	2.503	2.530	2.534	2.573	2.596	2.550
	Ebn at Noon		849	892	926	914	919	920	915	919	908	877	852	818
	Edh at Noon		74	86	93	111	108	106	105	100	94	82	70	70
(44) All-Sky Solar Radiation	RadAvg		2.11	3.00	4.29	5.73	6.75	7.47	6.90	5.96	5.14	3.72	2.47	1.79
	RadStd		0.32	0.42	0.39	0.44	0.49	0.40	0.39	0.32	0.31	0.27	0.23	0.30

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	-1.03	N/A	N/A	N/A	N/A	N/A	+123	N/A

Nomenclature: See separate page